

Quadratics & Polynomials

ANSWER KEY



Factoring quadratics

- 1 Solve $x^2 + 7x + 12 = 0$ by factoring.

$$(x + 3)(x + 4) = 0, \text{ so } x = -3 \text{ or } x = -4.$$

- 2 Solve $x^2 - 9x + 20 = 0$ by factoring.

$$(x - 4)(x - 5) = 0, \text{ so } x = 4 \text{ or } x = 5.$$

- 3 Solve $2x^2 + 5x - 3 = 0$ by factoring.

$$(2x - 1)(x + 3) = 0, \text{ so } x = \frac{1}{2} \text{ or } x = -3.$$

Vertex form and completing the square

- 4 Write $y = x^2 - 6x + 5$ in vertex form by completing the square, and state the vertex.

$$x^2 - 6x + 5 = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4. \text{ Vertex: } (3, -4).$$

- 5 Solve $x^2 + 8x + 3 = 0$ by completing the square.

$$x^2 + 8x = -3, \text{ so } (x + 4)^2 - 16 = -3, \text{ giving } (x + 4)^2 = 13, \text{ so } x = -4 \pm \sqrt{13}.$$

The quadratic formula and discriminant

- 6 Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

$$a = 3, b = -5, c = -2. \text{ Discriminant} = 25 + 24 = 49. x = \frac{5 \pm 7}{6}, \text{ so } x = 2 \text{ or } x = -\frac{1}{3}.$$

- 7 Find the discriminant of $2x^2 + 3x + 5 = 0$ and state the nature of its roots.

$$\text{Discriminant} = 3^2 - 4(2)(5) = 9 - 40 = -31, \text{ which is negative, so the equation has no real roots.}$$

- 8 For what values of k does $x^2 + kx + 16 = 0$ have exactly one repeated real root?

$$\text{A repeated root occurs when the discriminant is zero: } k^2 - 4(1)(16) = 0, \text{ so } k^2 = 64, \text{ giving } k = \pm 8.$$

Graphing parabolas

- 9 The graph of $y = x^2 - 2x - 3$ is shown below. Use the graph to state the x -intercepts, then confirm your answer algebraically by factoring.

From the graph, the parabola crosses the x -axis at $x = -1$ and $x = 3$. Algebraically:

$$x^2 - 2x - 3 = (x - 3)(x + 1) = 0, \text{ so } x = 3 \text{ or } x = -1.$$

- 10 The graph of $y = -2x^2 + 8x - 3$ opens which direction, and what is the x -coordinate of its vertex?

Since $a = -2 < 0$, the parabola opens downward. Vertex: $x = -\frac{b}{2a} = -\frac{8}{2(-2)} = 2$.

Polynomial multiplication and special products

- 11 Expand $(2x - 5)^2$.

$$(2x - 5)^2 = 4x^2 - 20x + 25.$$

- 12 Multiply $(x + 7)(x - 7)$.

Using the difference-of-squares identity: $x^2 - 49$.

- 13 Multiply $(3x + 4)(3x - 4)$.

$$9x^2 - 16.$$

Polynomial division and the remainder theorem

- 14 Use synthetic division to divide $x^3 - 2x^2 - 5x + 6$ by $(x - 1)$, and state the quotient and remainder.

Using synthetic division with root 1 on coefficients 1, -2, -5, 6: bring down 1; $1(1) = 1$, $-2 + 1 = -1$; $-1(1) = -1$, $-5 + (-1) = -6$; $-6(1) = -6$, $6 + (-6) = 0$. Quotient: $x^2 - x - 6$, remainder: 0.

- 15 Use the Remainder Theorem to find the remainder when $x^3 + 4x^2 - x + 2$ is divided by $(x + 2)$.

By the Remainder Theorem, the remainder equals $f(-2)$:

$$(-2)^3 + 4(-2)^2 - (-2) + 2 = -8 + 16 + 2 + 2 = 12.$$

Quadratic word problems

- 16 A ball is thrown upward so that its height is modeled by $h(t) = -16t^2 + 64t + 5$ (in feet), where t is time in seconds. Find the time at which the ball reaches its maximum height, and find that maximum height.

Maximum occurs at $t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$ seconds.

$$h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69 \text{ feet.}$$

- 17 The area of a rectangle is $x^2 + 2x - 15$ square meters, and its width is $x - 3$. Find its length in terms of x by factoring.

$$x^2 + 2x - 15 = (x + 5)(x - 3). \text{ Since width} = x - 3, \text{ the length is } x + 5.$$

Sum and product of roots

- 18 For $x^2 - 7x + 10 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

$$\text{Sum} = -\frac{-7}{1} = 7. \text{ Product} = \frac{10}{1} = 10.$$

- 19 A quadratic equation has roots 3 and -5 . Write the equation in standard form.

$$(x - 3)(x + 5) = x^2 + 2x - 15 = 0.$$

A well-designed word problem: maximizing weekly profit

- 20 This question has three parts. A company's weekly profit (in thousands of dollars) from producing x hundred units is modeled by $P(x) = -2x^2 + 20x - 32$. (a) Factor $P(x)$ to find the break-even production levels, where profit is zero. (b) Find the vertex of the parabola to determine the production level that maximizes profit, and state the maximum profit. (c) State the range of production levels (in hundreds of units) for which profit is positive.

(a) $P(x) = -2(x^2 - 10x + 16) = -2(x - 2)(x - 8)$. Break-even at $x = 2$ and $x = 8$ (hundreds of units).

(b) Vertex: $x = -\frac{20}{2(-2)} = 5$. $P(5) = -2(25) + 20(5) - 32 = -50 + 100 - 32 = 18$, so the maximum profit is 18 thousand dollars at $x = 5$ hundred units (500 units). (c) Profit is positive between the roots: $2 < x < 8$, i.e. between 200 and 800 units.

Multiple choice — easy

- 21 Solve $x^2 - 15x + 54 = 0$ by factoring.

- a. 6, 8
- b. 7, 9
- c. 6, 9 ✓
- d. $-6, -9$

$$(x - 9)(x - 6) = 0, \text{ so } x = 9 \text{ or } x = 6.$$

22 Solve $x^2 - 6x = 0$ by factoring.

- a. 1, 6
- b. 0, 5
- c. 0, - 6
- d. 0, 6 ✓

$(x - 0)(x - 6) = 0$, so $x = 0$ or $x = 6$.

23 Solve $x^2 + 5x + 4 = 0$ by factoring.

- a. 4, 1
- b. -4, - 2
- c. -3, - 1
- d. -4, - 1 ✓

$(x - (-1))(x - (-4)) = 0$, so $x = -1$ or $x = -4$.

24 Solve $x^2 - 9 = 0$ by factoring.

- a. -3, 3 ✓
- b. 3, - 3
- c. -3, 2
- d. -2, 3

$(x - (-3))(x - 3) = 0$, so $x = -3$ or $x = 3$.

25 Solve $x^2 + 4x = 0$ by factoring.

- a. 4, 0
- b. -3, 0
- c. -4, 0 ✓
- d. -4, - 1

$(x - 0)(x - (-4)) = 0$, so $x = 0$ or $x = -4$.

26 Solve $x^2 - 2x = 0$ by factoring.

a. 0, 2 ✓

b. 0, -2

c. 0, 1

d. 1, 2

$(x - 0)(x - 2) = 0$, so $x = 0$ or $x = 2$.

27 Solve $x^2 + 3x + 2 = 0$ by factoring.

a. 2, 1

b. -1, -1

c. -2, -1 ✓

d. -2, -2

$(x - (-2))(x - (-1)) = 0$, so $x = -2$ or $x = -1$.

28 Solve $x^2 - 2x - 63 = 0$ by factoring.

a. -6, 9

b. -7, 8

c. 7, -9

d. -7, 9 ✓

$(x - 9)(x - (-7)) = 0$, so $x = 9$ or $x = -7$.

29 Solve $x^2 - 4x - 21 = 0$ by factoring.

a. $-2, 7$

b. $-3, 7$ ✓

c. $3, -7$

d. $-3, 6$

$(x - 7)(x - (-3)) = 0$, so $x = 7$ or $x = -3$.

30 Solve $x^2 - x - 12 = 0$ by factoring.

a. $-3, 4$ ✓

b. $-2, 4$

c. $-3, 3$

d. $3, -4$

$(x - (-3))(x - 4) = 0$, so $x = -3$ or $x = 4$.

31 Solve $x^2 - 13x + 42 = 0$ by factoring.

a. $6, 6$

b. $6, 7$ ✓

c. $7, 7$

d. $-6, -7$

$(x - 6)(x - 7) = 0$, so $x = 6$ or $x = 7$.

32 Solve $x^2 + 11x + 28 = 0$ by factoring.

a. $-7, -4$ ✓

b. $-6, -4$

c. $-7, -5$

d. $7, 4$

$(x - (-4))(x - (-7)) = 0$, so $x = -4$ or $x = -7$.

33 Solve $x^2 + 12x + 35 = 0$ by factoring.

a. $-6, -5$

b. $7, 5$

c. $-7, -5$ ✓

d. $-7, -6$

$(x - (-5))(x - (-7)) = 0$, so $x = -5$ or $x = -7$.

34 Solve $x^2 - 9x = 0$ by factoring.

a. $0, 8$

b. $0, -9$

c. $0, 9$ ✓

d. $1, 9$

$(x - 0)(x - 9) = 0$, so $x = 0$ or $x = 9$.

35 Solve $x^2 - 5x - 14 = 0$ by factoring.

a. $-2, 7$ ✓

b. $-2, 6$

c. $2, -7$

d. $-1, 7$

$(x - (-2))(x - 7) = 0$, so $x = -2$ or $x = 7$.

Multiple choice — medium

36 Solve $2x^2 - 17x + 30 = 0$ using the quadratic formula.

a. $\frac{5}{2}, 5$

b. $\frac{7}{2}, 6$

c. $\frac{5}{2}, 6$ ✓

d. $-\frac{5}{2}, -6$

$a = 2, b = -17, c = 30$. Discriminant $= (-17)^2 - 4(2)(30) = 289 - 240 = 49$. $x = \frac{17 \pm 7}{4}$, so $x = 6$ or $x = \frac{5}{2}$.

37 Solve $2x^2 + 8x - 24 = 0$ using the quadratic formula.

a. $-6, 1$

b. $6, -2$

c. $-5, 2$

d. $-6, 2$ ✓

$a = 2, b = 8, c = -24$. Discriminant $= (8)^2 - 4(2)(-24) = 64 + 192 = 256$. $x = \frac{-8 \pm 16}{4}$, so $x = 2$ or $x = -6$.

38 Solve $3x^2 - 9x - 30 = 0$ using the quadratic formula.

a. $-2, 5$ ✓

b. $-1, 5$

c. $-2, 4$

d. $2, -5$

$a = 3, b = -9, c = -30$. Discriminant $= (-9)^2 - 4(3)(-30) = 81 + 360 = 441$. $x = \frac{9 \pm 21}{6}$, so $x = 5$ or $x = -2$.

39 Solve $x^2 - 8x + 12 = 0$ using the quadratic formula.

a. $-2, -6$

b. $3, 6$

c. $2, 5$

d. $2, 6$ ✓

$a = 1, b = -8, c = 12$. Discriminant $= (-8)^2 - 4(1)(12) = 64 - 48 = 16$. $x = \frac{8 \pm 4}{2}$, so $x = 6$ or $x = 2$.

40 Solve $2x^2 - 20x + 48 = 0$ using the quadratic formula.

a. $4, 5$

b. $4, 6$ ✓

c. $5, 6$

d. $-4, -6$

$a = 2, b = -20, c = 48$. Discriminant $= (-20)^2 - 4(2)(48) = 400 - 384 = 16$. $x = \frac{20 \pm 4}{4}$, so $x = 6$ or $x = 4$.

41 Solve $2x^2 - 6x - 8 = 0$ using the quadratic formula.

a. $-1, 4$ ✓

b. $-1, 3$

c. $0, 4$

d. $1, -4$

$a = 2, b = -6, c = -8$. Discriminant $= (-6)^2 - 4(2)(-8) = 36 + 64 = 100$. $x = \frac{6 \pm 10}{4}$, so $x = 4$ or $x = -1$.

42 Solve $3x^2 + 3x - 60 = 0$ using the quadratic formula.

a. 5, -4

b. -5, 4 ✓

c. -5, 3

d. -4, 4

$a = 3, b = 3, c = -60$. Discriminant $= (3)^2 - 4(3)(-60) = 9 + 720 = 729$. $x = \frac{-3 \pm \sqrt{729}}{6}$, so $x = 4$ or $x = -5$.

43 Solve $x^2 - 3x - 10 = 0$ using the quadratic formula.

a. -2, 5 ✓

b. -2, 4

c. -1, 5

d. 2, -5

$a = 1, b = -3, c = -10$. Discriminant $= (-3)^2 - 4(1)(-10) = 9 + 40 = 49$. $x = \frac{3 \pm \sqrt{49}}{2}$, so $x = 5$ or $x = -2$.

44 Solve $2x^2 - 12x = 0$ using the quadratic formula.

a. 0, -6

b. 0, 5

c. 1, 6

d. 0, 6 ✓

$a = 2, b = -12, c = 0$. Discriminant $= (-12)^2 - 4(2)(0) = 144 = 144$. $x = \frac{12 \pm \sqrt{144}}{4}$, so $x = 6$ or $x = 0$.

45 Solve $2x^2 + 6x - 8 = 0$ using the quadratic formula.

a. $-4, 0$

b. $-4, 1$ ✓

c. $4, -1$

d. $-3, 1$

$a = 2, b = 6, c = -8$. Discriminant $= (6)^2 - 4(2)(-8) = 36 + 64 = 100$. $x = \frac{-6 \pm 10}{4}$, so $x = 1$ or $x = -4$.

46 Solve $2x^2 - 6x = 0$ using the quadratic formula.

a. $0, 2$

b. $0, 3$ ✓

c. $0, -3$

d. $1, 3$

$a = 2, b = -6, c = 0$. Discriminant $= (-6)^2 - 4(2)(0) = 36 = 36$. $x = \frac{6 \pm 6}{4}$, so $x = 3$ or $x = 0$.

47 Solve $x^2 - x - 2 = 0$ using the quadratic formula.

a. $-1, 1$

b. $-1, 2$ ✓

c. $1, -2$

d. $0, 2$

$a = 1, b = -1, c = -2$. Discriminant $= (-1)^2 - 4(1)(-2) = 1 + 8 = 9$. $x = \frac{1 \pm 3}{2}$, so $x = 2$ or $x = -1$.

48 Solve $3x^2 + 12x + 9 = 0$ using the quadratic formula.

a. $-3, -1$ ✓

b. $-2, -1$

c. $-3, -2$

d. $3, 1$

$a = 3, b = 12, c = 9$. Discriminant $= (12)^2 - 4(3)(9) = 144 - 108 = 36$. $x = \frac{-12 \pm 6}{6}$, so $x = -1$ or $x = -3$.

49 Solve $x^2 - x - 30 = 0$ using the quadratic formula.

a. $-5, 5$

b. $5, -6$

c. $-4, 6$

d. $-5, 6$ ✓

$a = 1, b = -1, c = -30$. Discriminant $= (-1)^2 - 4(1)(-30) = 1 + 120 = 121$. $x = \frac{1 \pm 11}{2}$, so $x = 6$ or $x = -5$.

50 Solve $2x^2 - 14x + 12 = 0$ using the quadratic formula.

a. $-1, -6$

b. $2, 6$

c. $1, 6$ ✓

d. $1, 5$

$a = 2, b = -14, c = 12$. Discriminant $= (-14)^2 - 4(2)(12) = 196 - 96 = 100$. $x = \frac{14 \pm 10}{4}$, so $x = 6$ or $x = 1$.

Multiple choice — hard

51 Find the vertex of the parabola $y = -2x^2 - 28x - 92$.

a. $(-7, 6)$ ✓

b. $(-5, 6)$

c. $(7, 6)$

d. $(-7, -6)$

$$h = -\frac{b}{2a} = -\frac{-28}{-4} = -7. k = y(-7) = -2(-7)^2 + 196 - 92 = 6. \text{ Vertex: } (-7, 6).$$

52 Find the vertex of the parabola $y = -x^2 + 16x - 71$.

a. $(10, -7)$

b. $(8, -7)$ ✓

c. $(8, 7)$

d. $(-8, -7)$

$$h = -\frac{b}{2a} = -\frac{16}{-2} = 8. k = y(8) = -1(8)^2 + 128 - 71 = -7. \text{ Vertex: } (8, -7).$$

53 Find the vertex of the parabola $y = 3x^2 - 6x + 18$.

a. $(1, 15)$ ✓

b. $(-1, 15)$

c. $(3, 15)$

d. $(1, -15)$

$$h = -\frac{b}{2a} = -\frac{-6}{6} = 1. k = y(1) = 3(1)^2 - 6 + 18 = 15. \text{ Vertex: } (1, 15).$$

54 Find the vertex of the parabola $y = 3x^2 - 42x + 147$.

a. $(-7, 0)$

b. $(9, 0)$

c. $(7, 2)$

d. $(7, 0)$ ✓

$$h = -\frac{b}{2a} = -\frac{-42}{6} = 7. k = y(7) = 3(7)^2 - 294 + 147 = 0. \text{ Vertex: } (7, 0).$$

55 Find the vertex of the parabola $y = -3x^2 - 24x - 47$.

a. $(-2, 1)$

b. $(-4, 1)$ ✓

c. $(-4, -1)$

d. $(4, 1)$

$$h = -\frac{b}{2a} = -\frac{-24}{-6} = -4. \quad k = y(-4) = -3(-4)^2 + 96 - 47 = 1. \quad \text{Vertex: } (-4, 1).$$

56 Multiply $(5x + 7)^2$.

a. $25x^2 + 70x - 49$

b. $25x^2 + 70x + 49$ ✓

c. $25x^2 + 49$

d. $25x^2 - 70x + 49$

$$(5x + 7)^2 = 25x^2 + 70x + 49.$$

57 Multiply $(2x + 11)^2$.

a. $4x^2 - 44x + 121$

b. $4x^2 + 44x - 121$

c. $4x^2 + 44x + 121$ ✓

d. $4x^2 + 121$

$$(2x + 11)^2 = 4x^2 + 44x + 121.$$

58 Multiply $(5x + 6)(5x - 6)$.

a. $25x^2 + 36$

b. $25x^2 + 60x - 36$

c. $25x^2 - 38$

d. $25x^2 - 36$ ✓

Using the difference-of-squares identity: $25x^2 - 36$.

59 Multiply $(4x + 9)^2$.

a. $16x^2 + 81$

b. $16x^2 + 72x + 81$ ✓

c. $16x^2 + 72x - 81$

d. $16x^2 - 72x + 81$

$(4x + 9)^2 = 16x^2 + 72x + 81$.

60 Multiply $(2x + 9)(2x - 9)$.

a. $4x^2 + 81$

b. $4x^2 - 81$ ✓

c. $4x^2 + 36x - 81$

d. $4x^2 - 83$

Using the difference-of-squares identity: $4x^2 - 81$.

- 61 For $2x^2 + 12x + 16 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $-6, 8$ ✓

b. $8, -6$

c. $6, 8$

d. $-6, -8$

$$\text{Sum} = -\frac{12}{2} = -6. \text{ Product} = \frac{16}{2} = 8.$$

- 62 For $2x^2 - 13x + 8 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $\frac{13}{2}, -4$

b. $-\frac{13}{2}, 4$

c. $\frac{13}{2}, 4$ ✓

d. $4, \frac{13}{2}$

$$\text{Sum} = -\frac{-13}{2} = \frac{13}{2}. \text{ Product} = \frac{8}{2} = 4.$$

- 63 For $2x^2 + 14x + 2 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $1, -7$

b. $-7, 1$ ✓

c. $7, 1$

d. $-7, -1$

$$\text{Sum} = -\frac{14}{2} = -7. \text{ Product} = \frac{2}{2} = 1.$$

- 64 For $x^2 + 6x + 6 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. 6, -6

b. 6, 6

c. -6, 6 ✓

d. -6, -6

$$\text{Sum} = -\frac{6}{1} = -6. \text{ Product} = \frac{6}{1} = 6.$$

- 65 For $x^2 + 13x + 7 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. -13, -7

b. -13, 7 ✓

c. 13, 7

d. 7, -13

$$\text{Sum} = -\frac{13}{1} = -13. \text{ Product} = \frac{7}{1} = 7.$$